GRIDSHEDWATCH

Scoring early warnings of grid supply emergencies against documented events

An open benchmark, scored the way weather services score their warnings

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THE PROBLEM

Many methods claim to see grid emergencies coming. No common set of real emergencies tests them.

Own events

Each study picks its own storms, often a single one, or simulated grids.

Per-hour metrics

They reward flagging many hours, not one alert in time. Point-adjusted F1 even ranks random scores above every real method.

False alarms unseen

An operator pays for every false alarm, but most evaluations never count them where nothing happened.

Weather services solved this decades ago: score warnings as events, by how many were caught, how early, and with how many false alarms.

Motivation: Winter Storm Uri (2021) cut power to 4.5 million people in Texas; Elliott (2022) forced load sheds at six Southeastern utilities

EVENTS

Every firm load shed of 250 MW or more for lack of supply, 2020–2026

Storm	Balancing authority	Load shed
Heat wave, Aug 2020	CAISO	1,000 MW
Uri, Feb 2021	ERCOT	20,000 MW
Uri, Feb 2021	SPP	2,718 MW
Uri, Feb 2021	MISO (MISO South)	700 MW
Elliott, Dec 2022	TVA	over 3,000 MW
Elliott, Dec 2022	LG&E and KU	317 MW
Elliott, Dec 2022	Duke Energy Carolinas	1,000 MW
Elliott, Dec 2022	Duke Energy Progress	961 MW

Complete under the rule; checked against NERC’s load-shed tables and DOE OE-417. Two EEA 3 emergencies without a shed are reported as case studies

DATA

48 grids of public hourly data, frozen, and five signals every method sees

48 grids

Every EIA-930 balancing authority with 90% coverage. 9 had an event; 39 are quiet.

Frozen

Demand, day-ahead forecast and net generation, 2019–2026, with checksums, because EIA revises past values.

5 signals

How far demand misses the operator’s own forecast, how much power the grid imports, and how unusual each is for the time of year.

2019 is for calibration only; scoring starts in January 2020. No weather, outage or price data: the benchmark asks what public grid totals alone can warn of.

Code MIT, data and results CC BY 4.0, Croissant metadata and datasheet included

SCORING

Alerts are scored as events: caught, how early, and with how many false alarms

Watch

Counts if it starts in the 7 days before onset.

Warning

Counts from 7 days before onset to 2 days after. Headline results use only warnings before onset.

Measured: events caught, lead time, false alarm ratio, and alerts per year on the 39 quiet grids.

Methods submit per-signal scores; shared rules turn every method’s scores into alerts the same way. Methods can also submit alert times directly

FAIR COMPARISON

Every method alerts at the same rate, set without looking at any event

Budget: 12 watches and 2 warnings per grid per year, on the median grid. About as often as operators already alert the public in a strained year.

Set on quiet grids: each method’s threshold is tuned on 19 grids that never had an emergency. No event grid is seen.

In practice: this is how any alarm threshold is set. Pick, from past data, the setting that would have alerted about once a month, then keep it fixed.

Without a matched budget, a comparison mostly rewards whoever alerts most often.

The other 20 quiet grids measure false-alarm burden; curves at ¼× to 4× the budget are in the appendix

REFERENCE METHODS

Eleven methods, from a rule an operator could write to a foundation model

Rules and statistics

- Operator-style rule: demand above its forecast for 3 hours
- Seasonal z-score: unusual for the time of year
- CUSUM: small, steady deviations that add up
- Critical slowing down: a system near a tipping point swings more and recovers slower

Forecasters

Predict the next hour and alert on large misses: LightGBM (gradient-boosted trees), LSTM, PatchTST (a transformer), Chronos-Bolt (a pretrained time-series model)

Detectors

Sub-PCA: patterns unlike past ones. HTM: learns sequences and flags surprises

Floor

Random scores: what pure luck achieves at the same alert rate

All see the same five signals; learned models fit on 2019 only; settings are published defaults, none tuned on events

RESULT 1

With data as replayed, four methods beat chance

Method	Watches (of 8)	By chance	Warnings before onset	Beats chance
LightGBM	7.2	2.2	6.0	Both
LSTM	6.2	1.9	4.0	Both
HTM	6.0	1.7	5.0	Both
PatchTST	6.0	1.8	5.6	Both
Chronos-Bolt	6.0	2.6	1.0	No
CUSUM	5.0	1.5	4.0	Warnings only
Seasonal z-score	5.0	0.9	3.0	No
Operator-style rule	5.0	2.3	2.0	No
Critical slowing down	4.0	2.3	1.0	No
Sub-PCA	2.0	0.8	0.0	No
Random scores	1.0	1.8	0.2	No
Forecast-anomaly control	**8.0**	1.3	3.0	Yes (weather)

“By chance” = the same alerts moved to random times in the same season, so a method can’t pass just by alerting a lot in winter. Listed by watches caught, not ranked: 8 events can’t separate the top methods. Last row: the operators’ own forecast, next slide

RESULT 2

The operators’ own forecast catches all 8; some methods are more selective

In Uri and Elliott, watches on…	grids that shed (7)	equally cold grids that coped (11)
Forecast-anomaly control	100%	100%
LightGBM	100%	44%
LSTM	89%	**25%**
HTM	83%	53%
PatchTST	74%	24%
Chronos-Bolt	86%	**9%**
Random scores	14%	16%

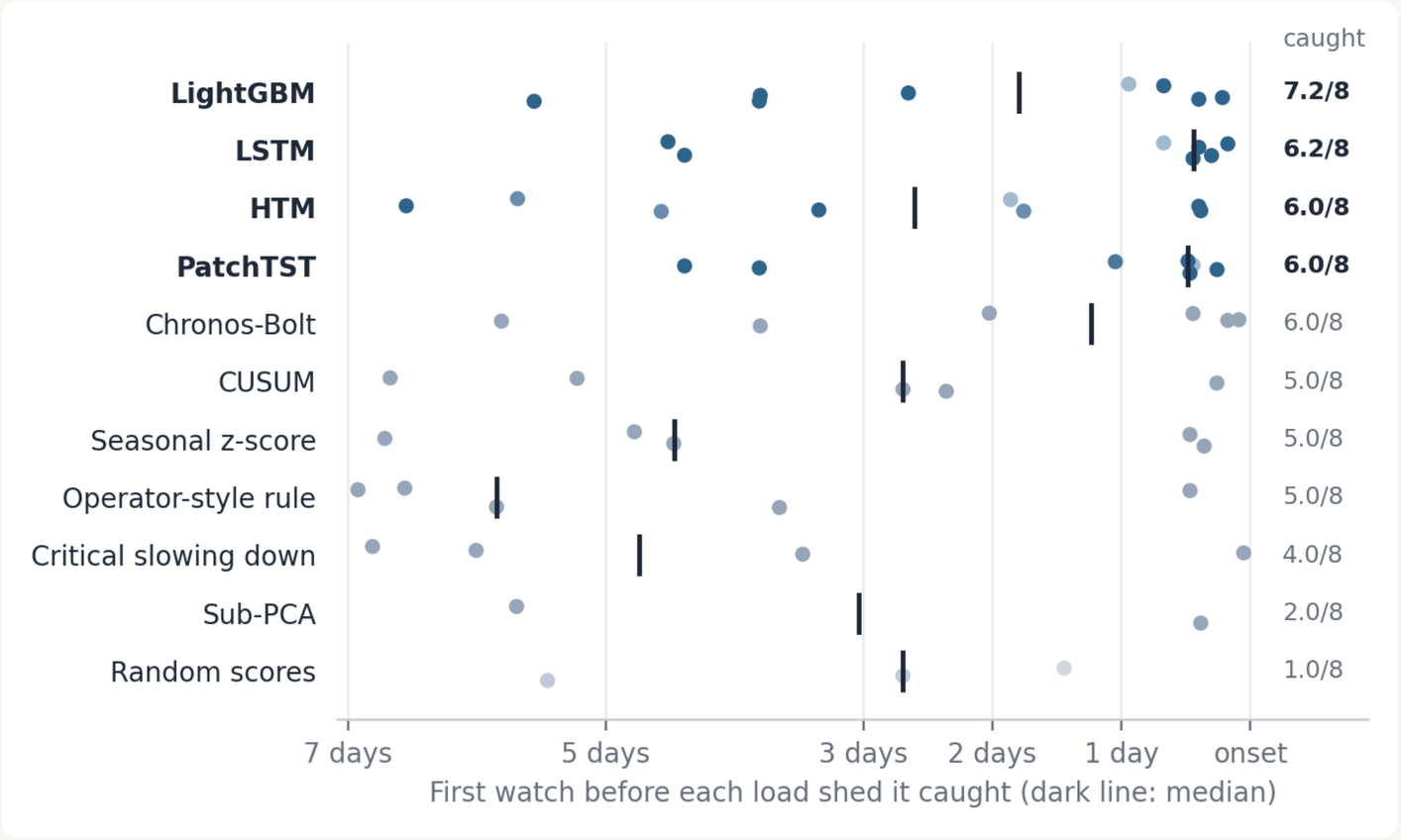
Alerting when the day-ahead forecast is unusual for the season watches every load shed, about 16 hours ahead.

But it can’t tell a grid that sheds from one that copes with the same cold. Telling those apart is where methods differ.

Bold: significant after correction (exact test, shuffling which grids shed within each storm). Few grids: suggestive, not settled

RESULT 3

With a day’s real-time delay, no method is shown to beat chance



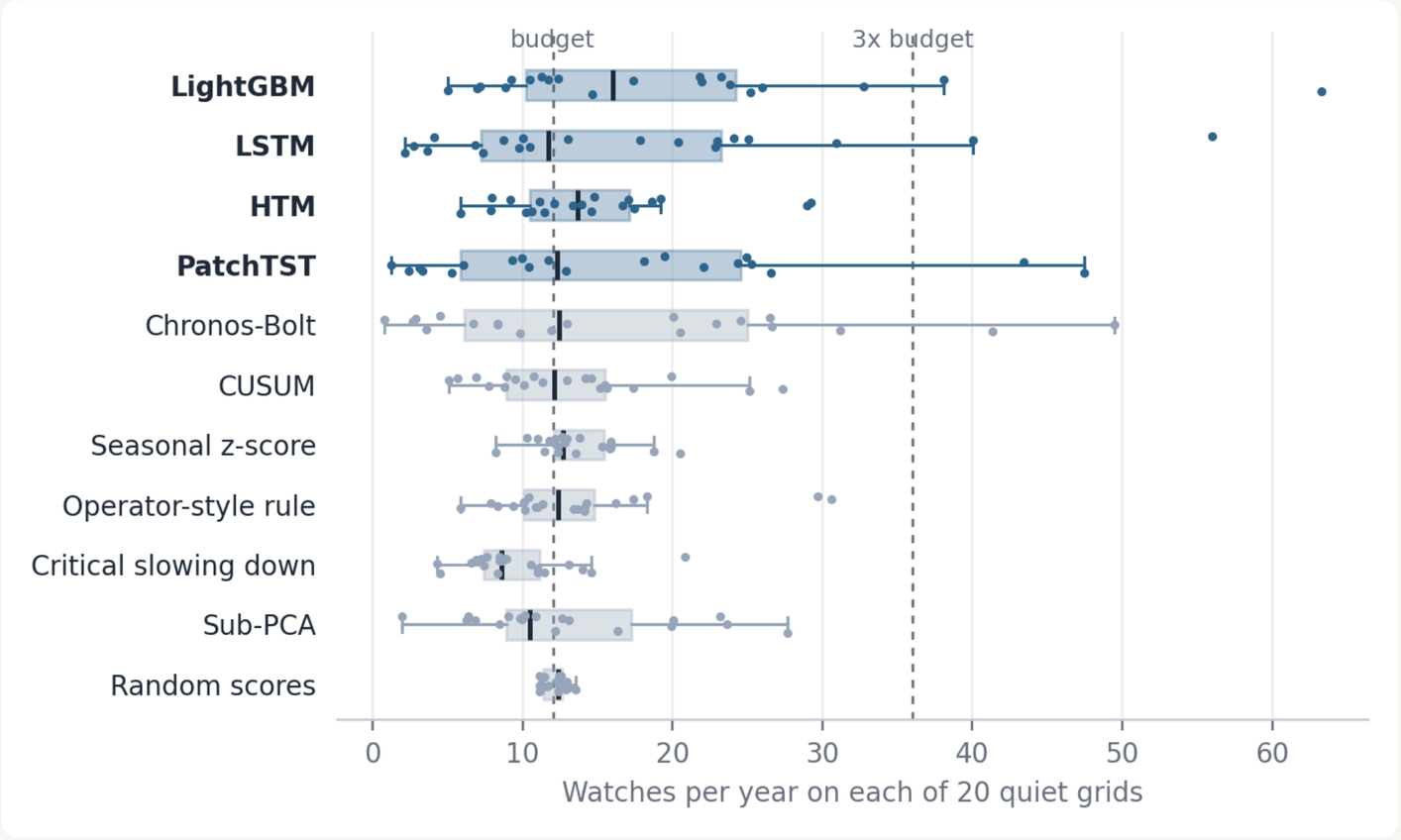
Each dot is a load shed caught; the line is the median.

EIA publishes net generation the next morning, so live alerts come 11–35 h late. A day late, HTM keeps 4.4 of 8 watches, LSTM 2.0: none beats chance after correction.

Bold: the four methods that beat chance. Paler dots: caught by fewer seeds. Descriptive, not a significance test

RESULT 4

Same typical alert rate; forecasters spike far more on some grids



20 quiet grids not used to set thresholds; box = middle half of grids, line = median

Each dot is a grid that never had an emergency.

Every method sits near the budget on a typical grid. The forecasters reach 47–63 watches a year on their worst grid; HTM and the rules stay under about 30.

LIMITS

What this benchmark does not claim

Few events, three storms

Evidence of real warning, not a detection rate to expect in operation. 7 of the 8 come from two storms.

Heat waves

California 2020 is the hardest event: demand tracked its forecast; the shortfall was on the supply side.

Public totals only

Hourly grid totals can’t see lost generation; net generation arrives the next morning.

Budget matched on quiet grids

Alert rates differ on the event grids, so raw counts aren’t compared across methods; chance tests are.

Not a replacement for an operator’s situational awareness

SUMMARY

A scorecard for grid early warning, and what it shows so far

1. Every documented load shed since 2020, scored like weather warnings at one alert budget
2. With data as replayed, four methods beat chance; eight events can’t rank them
3. The operators’ own forecast catches all eight but can’t tell a cold week from a shortfall; some methods can, a little
4. Open: code, data and results; new methods and new events welcome